

CSCI 332, Fall 2025

In-class Activity

For each of the three problems, answer the questions at the bottom of the sheet.

1. You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?
2. You are given an integer array `cost` where `cost[i]` is the cost of i th step on a staircase. Once you pay the cost, you can either climb one or two steps. You can either start from the step with index 0, or the step with index 1. Return the minimum cost to reach the top of the staircase.
3. In your country, there are only coins of denominations 1, 3, and 4. Given a nonnegative integer n , what is the fewest number of coins you can use to represent n ?

Questions to answer for each problem:

1. Write the definition of the recursive subproblems you will solve. Make sure to give these a descriptive name.
2. Write the recurrence to solve your recursive subproblems.
3. In what direction will you fill your memoization array?
4. Write the pseudocode for the memoized algorithm.